SOFTWARE TEST PLAN

Tutoring Platform v1.0 – Group 43

DOCUMENT INFORMATION	
Version	1.0
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Prepared By	Group 43
Reviewed By	
Preparation Date	April 2026
System Under Test	Tutoring Platform v1.0
Test Environment	React + Node.js + Express + PostgreSQL
Methodology	COMET / IEEE 29119

In Scope	Out of Scope
User Registration & Authentication · Tutor Profile Management · Tutor Search & Filtering · Session Booking & Management · Booking-Based Messaging · Review & Rating System · Tutor Recommendation Engine · Learning Path Recommendation · Role-Based Access Control · REST API Contract · Negative / Error-path Testing · Boundary Value Analysis · Security Testing	Native mobile app testing · Load testing beyond 100 concurrent users · Multi-language UI · Real payment gateway integration · Third-party video call quality testing · Email delivery testing

Test Types
Functional · Negative / Error-path · Boundary Value · Security · Integration (API contract) · Role-Based Access Control

STATUS LEGEND & TEST ID CONVENTION

🔴 FAIL 🟡 PARTIAL 🟢 PASS ⬜ NOT EXEC 🔵 BLOCKED

Test ID Convention: TC-[MODULE]-[NUMBER] e.g., TC-AUTH-01

Req ID	Requirement Summary	Test Sheet	Priority	Test Count
F1	User Registration	UC1 – Auth & Reg	P1 – Critical	6
F2	User Authentication	UC1 – Auth & Reg	P1 – Critical	6
F3	Tutor Profile Management	UC2 – Profile	P1 – Critical	5
F4	Tutor Search & Filters	UC3 – Search	P1 – Critical	6
F5	Tutor Profile Viewing	UC3 – Search	P2 – High	4
F6	Session Booking	UC4 – Booking	P1 – Critical	7
F7	Booking Management (states)	UC4 – Booking	P1 – Critical	5
F8	Messaging System (booking-based)	UC5 – Messaging	P2 – High	5

F9	Review & Rating System	UC6 – Review	P2 – High	5
F10	Tutor Recommendation System	UC7 – Recommend	P2 – High	5
F11	Learning Path Recommendation	UC7 – Recommend	P2 – High	5

TOTAL TEST CASES ACROSS ALL SHEETS: 59

UC1 – AUTHENTICATION & REGISTRATION TEST CASES

Module: AUTH | Registration, Login, Role-Based Redirect, Session & Security | Maps to: F1, F2

TC ID	Test Category	Test Scenario	Precondition	Test Steps	Expected Result	Actual Result	Req. Mapped	Priority	Status	Defect ID
TC-AUTH-01	Functional	Valid student registration	No existing account with same email	1. Go to Register page 2. Select 'Student' role 3. Fill name, university email, password 4. Click Register	Account created; redirected to Student Dashboard	Account created; redirected to Student Dashboard	F1	P1 – Critical	Pass	
TC-AUTH-02	Functional	Valid tutor registration	No existing account with same email	1. Go to Register page 2. Select 'Tutor' role 3. Fill name, email, subjects, password 4. Click Register	Account created; redirected to Tutor Dashboard	Account created; redirected to Tutor Dashboard	F1	P1 – Critical	Pass	
TC-AUTH-03	Negative	Duplicate email registration	User with email already exists	1. Go to Register page 2. Enter already-used email 3. Fill remaining fields 4. Click Register	Error: 'Email already registered' shown; no account created	Error: 'Email already registered' shown; no account created	F1	P1 – Critical	Pass	
TC-AUTH-04	Functional	Valid student login	Student account exists in DB	1. Go to Login page 2. Enter registered email 3. Enter correct password 4. Click Sign In	Redirected to Student Dashboard	Redirected to Student Dashboard	F2	P1 – Critical	Pass	
TC-AUTH-05	Functional	Valid tutor login	Tutor account exists in DB	1. Go to Login page 2. Enter tutor email 3. Enter correct password 4. Click Sign In	Redirected to Tutor Dashboard	Redirected to Tutor Dashboard	F2	P1 – Critical	Pass	
TC-AUTH-06	Negative	Invalid password – correct email	User exists in DB	1. Enter valid email 2. Enter wrong password 3. Click Sign In	Error: 'Invalid credentials' shown	Error: 'Invalid credentials' shown	F2	P1 – Critical	Pass	
TC-AUTH-07	Negative	Non-existent user login	No such user in DB	1. Enter unregistered email 2. Enter any password 3. Click Sign In	Error: 'Invalid credentials' shown	Error: 'Invalid credentials' shown	F2	P1 – Critical	Pass	
TC-AUTH-08	Negative	Empty credentials submission	Login page loaded	1. Leave all fields blank 2. Click Sign In	Error message shown; no redirect	Error message shown; no redirect	F2	P2 – High	Pass	
TC-AUTH-09	Security	SQL injection in login fields	Login page loaded	1. Enter ' OR 1=1 -- in email field 2. Enter any password 3. Click Sign In	Login rejected; error shown; no data exposed	Login rejected; error shown; no data exposed	F2	P1 – Critical	Pass	

TC-AUTH-10	Negative	Password below minimum length	Registration page loaded	1. Enter valid email 2. Enter password with 4 chars 3. Click Register	Error: 'Password must be at least 8 characters'	Error: 'Password must be at least 8 characters'	F1	P2 – High	Pass	
TC-AUTH-11	Role-Based	Student cannot access Tutor Dashboard	Student logged in	1. Log in as student 2. Manually navigate to /tutor/dashboard URL	Access denied; redirected to Student Dashboard	Access denied; redirected to Student Dashboard	F2	P1 – Critical	Pass	
TC-AUTH-12	Role-Based	Tutor cannot access Student booking flow	Tutor logged in	1. Log in as tutor 2. Attempt to access /student/book URL directly	Access denied; redirected to Tutor Dashboard	Access denied; redirected to Tutor Dashboard	F2	P1 – Critical	Pass	
Total Test Cases in this Sheet: 12										

UC3 – TUTOR SEARCH TEST CASES

Module: SEARCH | Filter by subject, price, rating, availability | Maps to: F4, F5

TC ID	Test Category	Test Scenario	Precondition	Test Steps	Expected Result	Actual Result	Req. Mapped	Priority	Status	Defect ID
TC-SRCH-01	Functional	Search tutor by subject	Student logged in; tutors with 'Math' in DB	1. Go to Search page 2. Enter 'Math' in subject filter 3. Click Search	List of tutors offering Math is displayed	List of tutors offering Math is displayed	F4	P1 – Critical	Pass	
TC-SRCH-02	Functional	Filter tutors by price range	Student logged in; tutors with varying prices in DB	1. Set price filter: Min ₹200 – Max ₹500 2. Click Search	Only tutors with hourly rate in ₹200–₹500 range displayed	Only tutors with hourly rate in ₹200–₹500 range displayed	F4	P1 – Critical	Pass	
TC-SRCH-03	Functional	Filter tutors by minimum rating	Student logged in; tutors with varying ratings in DB	1. Set rating filter to 4+ 2. Click Search	Only tutors with rating ≥ 4.0 displayed	Only tutors with rating ≥ 4.0 displayed	F4	P1 – Critical	Pass	
TC-SRCH-04	Functional	Filter tutors by availability	Student logged in	1. Select 'Monday' and '10:00 AM' in availability filter 2. Click Search	Only tutors available on Monday at 10:00 AM shown	Only tutors available on Monday at 10:00 AM shown	F4	P1 – Critical	Pass	
TC-SRCH-05	Negative	Search with no matching tutor	No tutor offers 'Quantum Computing' in DB	1. Enter 'Quantum Computing' in subject filter 2. Click Search	Message: 'No tutors found for this subject'	Message: 'No tutors found for this subject'	F4	P2 – High	Pass	
TC-SRCH-06	Functional	View tutor profile from search results	Search results populated	1. Perform a search 2. Click on a tutor's name/card in results	Tutor profile page loads with full details (subjects, rating, experience, availability)	Tutor profile page loads with full details (subjects, rating, experience, availability)	F5	P1 – Critical	Pass	
TC-SRCH-07	Boundary	Search with combined filters	Tutors with various subjects and prices in DB	1. Set subject = 'Physics', rating = 3+, price max = ₹400 2. Click Search	Results satisfy all three filter conditions simultaneously	Results satisfy all three filter conditions simultaneously	F4	P2 – High	Pass	
TC-SRCH-08	Negative	Search with empty subject field	Student logged in	1. Leave subject field blank 2. Click Search	All available tutors displayed OR validation message shown	All available tutors displayed OR validation message shown	F4	P2 – High	Pass	
TC-SRCH-09	Integration	Search results reflect real-time tutor availability	Tutor marks themselves unavailable	1. Tutor sets availability to 'Unavailable' 2. Student searches for that tutor's subject	Tutor does not appear in search results	Tutor does not appear in search results	F4, F3	P1 – Critical	Pass	
TC-SRCH-10	Functional	View tutor availability slots on profile	Tutor has weekly schedule configured	1. Open tutor profile 2. Check availability section	Weekly schedule with available time slots rendered correctly	Weekly schedule with available time slots rendered correctly	F5	P2 – High	Pass	

Total Test Cases in this Sheet: 10

UC2 – TUTOR PROFILE MANAGEMENT TEST CASES

Module: PROFILE | Create, Edit, View Tutor Profile | Maps to: F3

TC ID	Test Category	Test Scenario	Precondition	Test Steps	Expected Result	Actual Result	Req. Mapped	Priority	Status	Defect ID
TC-PROF-01	Functional	Tutor creates profile after registration	Tutor registered and logged in; no profile created yet	1. Go to Profile Setup page 2. Enter subjects, experience, hourly rate, bio 3. Set weekly availability 4. Click Save	Profile saved; visible in search results	Profile saved; visible in search results	F3	P1 – Critical	Pass	
TC-PROF-02	Functional	Tutor edits existing profile	Tutor logged in with existing profile	1. Go to Edit Profile page 2. Update hourly rate from ₹300 to ₹400 3. Click Save	Updated rate reflected immediately in profile and search results	Updated rate reflected immediately in profile and search results	F3	P1 – Critical	Pass	
TC-PROF-03	Negative	Tutor saves profile without mandatory fields	Tutor logged in	1. Go to Profile Setup page 2. Leave 'Subjects' field empty 3. Click Save	Error: 'Subjects field is required'	Error: 'Subjects field is required'	F3	P1 – Critical	Pass	
TC-PROF-04	Boundary	Tutor enters hourly rate of 0	Tutor on profile edit page	1. Set hourly rate to 0 2. Click Save	Error: 'Rate must be greater than 0' OR system accepts free tutoring with warning	Error: 'Rate must be greater than 0'	F3	P2 – High	Pass	
TC-PROF-05	Role-Based	Student cannot access tutor profile edit page	Student logged in	1. Log in as student 2. Navigate directly to /tutor/profile/edit URL	Access denied; redirected to Student Dashboard	Access denied; redirected to Student Dashboard	F3	P1 – Critical	Pass	
TC-PROF-06	Functional	Student views tutor public profile	Student logged in; tutor profile exists	1. Click on tutor from search results 2. Observe profile page	Read-only profile shown with subjects, rating, reviews, availability; no edit controls visible	Read-only profile shown with subjects, rating, reviews, availability; no edit controls visible	F3, F5	P1 – Critical	Pass	
Total Test Cases in this Sheet: 6										

UC4 – SESSION BOOKING & MANAGEMENT TEST CASES

Module: BOOKING | Request, Confirm, Cancel, State Transitions | Maps to: F6, F7

TC ID	Test Category	Test Scenario	Precondition	Test Steps	Expected Result	Actual Result	Req. Mapped	Priority	Status	Defect ID
TC-BOOK-01	Functional	Student books a session — valid slot	Student logged in; tutor available at chosen time	1. Open tutor profile 2. Select an available date and time slot 3. Click 'Request Session'	Booking created in PENDING state; tutor notified	Booking created in PENDING state; tutor notified	F6	P1 – Critical	Pass	
TC-BOOK-02	Functional	Tutor accepts a pending booking	Tutor logged in; pending booking exists	1. Go to Booking Requests tab 2. Click 'Accept' on pending request	Booking state changes to CONFIRMED; student notified	Booking state changes to CONFIRMED; student notified	F7	P1 – Critical	Pass	
TC-BOOK-03	Functional	Tutor rejects a pending booking	Tutor logged in; pending booking exists	1. Go to Booking Requests tab 2. Click 'Reject' on pending request	Booking state changes to CANCELLED; student notified	Booking state changes to CANCELLED; student notified	F7	P1 – Critical	Pass	
TC-BOOK-04	Negative	Student books an already-taken slot	Tutor has CONFIRMED booking at Mon 10:00 AM	1. Select Mon 10:00 AM slot for same tutor 2. Click 'Request Session'	Error: 'This slot is already booked'	Error: 'This slot is already booked'	F6, F7	P1 – Critical	Pass	
TC-BOOK-05	Functional	Student cancels a confirmed booking	Confirmed booking exists for student	1. Go to My Bookings 2. Click 'Cancel' on a CONFIRMED booking	Booking state changes to CANCELLED; slot freed for others	Booking state changes to CANCELLED; slot freed for others	F7	P1 – Critical	Pass	
TC-BOOK-06	Functional	Session marked as completed	Confirmed booking whose time has passed	1. Student/Tutor manually marks session as completed 2. Observe booking state	Booking state changes to COMPLETED; review prompt shown to student	Booking state changes to COMPLETED; review prompt shown to student	F7	P1 – Critical	Pass	
TC-BOOK-07	Negative	Booking with no tutor availability	Tutor has no open slots	1. Open tutor profile 2. Try to select a slot	Message: 'No available slots. Check back later.'	Message: 'No available slots. Check back later.'	F6	P2 – High	Pass	
TC-BOOK-08	Integration	Booking state visible consistently to both parties	Confirmed booking exists	1. Student views booking as CONFIRMED 2. Tutor views same booking	Both student and tutor see identical state CONFIRMED	Both student and tutor see identical state CONFIRMED	F6, F7	P1 – Critical	Pass	
TC-BOOK-09	Boundary	Student attempts to book past date	Booking form loaded	1. Set session date to yesterday's date 2. Click 'Request Session'	Error: 'Cannot book sessions in the past'	Error: 'Cannot book sessions in the past'	F6	P1 – Critical	Pass	
TC-BOOK-10	Role-Based	Tutor cannot initiate a booking request	Tutor logged in	1. Log in as tutor 2. Navigate to a student profile 3. Try to find booking button	No 'Request Session' button visible; booking initiation restricted to students	No 'Request Session' button visible; booking initiation restricted to students	F6	P2 – High	Pass	

Total Test Cases in this Sheet: 10

UC5 – MESSAGING SYSTEM TEST CASES

Module: MESSAGING | Booking-Based Threads | Maps to: F8

TC ID	Test Category	Test Scenario	Precondition	Test Steps	Expected Result	Actual Result	Req. Mapped	Priority	Status	Defect ID
TC-MSG-01	Functional	Student sends message to tutor within booking thread	Confirmed booking exists between student and tutor	1. Go to My Bookings 2. Open booking thread 3. Type message 4. Click Send	Message delivered and visible in thread for both parties	Message delivered and visible in thread for both parties	F8	P1 – Critical	Pass	
TC-MSG-02	Functional	Tutor replies in booking thread	Confirmed booking with existing student message	1. Tutor opens booking thread 2. Types reply 3. Clicks Send	Reply visible in thread; student sees new message notification	Reply visible in thread; student sees new message notification	F8	P1 – Critical	Pass	
TC-MSG-03	Negative	Messaging without an active booking	Student logged in; no booking with target tutor	1. Open tutor profile 2. Look for a direct message option outside booking	No messaging option available; message: 'You must have an active booking to message this tutor'	No messaging option available; message shown	F8	P1 – Critical	Pass	
TC-MSG-04	Negative	Send empty message	Booking thread open	1. Leave message field blank 2. Click Send	Message not sent; validation error shown	Message not sent; validation error shown	F8	P2 – High	Pass	
TC-MSG-05	Security	Send XSS payload in message	Booking thread open	1. Type <script>alert('XSS') </script> in message field 2. Click Send	Message stored and displayed as plain text; no script executed	Message stored and displayed as plain text; no script executed	F8	P1 – Critical	Pass	
Total Test Cases in this Sheet: 5										

UC6 – REVIEW & RATING TEST CASES

Module: REVIEW | Post-session ratings and feedback | Maps to: F9

TC ID	Test Category	Test Scenario	Precondition	Test Steps	Expected Result	Actual Result	Req. Mapped	Priority	Status	Defect ID
TC-REV-01	Functional	Student submits review after completed session	Session marked COMPLETED	1. Go to Completed Bookings 2. Click 'Leave Review' 3. Select 5-star rating 4. Type feedback text 5. Submit	Review saved; tutor's average rating updated; review visible on tutor profile	Review saved; tutor's average rating updated; review visible on tutor profile	F9	P1 – Critical	Pass	
TC-REV-02	Negative	Student attempts to review pending/active booking	Booking in CONFIRMED state	1. Go to My Bookings 2. Open CONFIRMED booking 3. Look for review option	No review option available; message: 'Reviews can only be submitted after session completion'	No review option visible	F9	P1 – Critical	Pass	
TC-REV-03	Boundary	Submit review with minimum rating (1 star)	Session marked COMPLETED	1. Select 1-star rating 2. Add feedback 3. Submit	1-star review saved; tutor average rating recalculated correctly	1-star review saved; tutor average rating recalculated	F9	P2 – High	Pass	
TC-REV-04	Negative	Submit duplicate review for same session	Student already reviewed this session	1. Go to completed booking already reviewed 2. Attempt to submit another review	Error: 'You have already reviewed this session'	Error: 'You have already reviewed this session'	F9	P1 – Critical	Pass	
TC-REV-05	Negative	Submit review without rating selection	Review form open	1. Write feedback text 2. Do not select star rating 3. Click Submit	Error: 'Rating is required'; form not submitted	Error: 'Rating is required'; form not submitted	F9	P2 – High	Pass	
Total Test Cases in this Sheet: 5										

UC7 – RECOMMENDATION SYSTEM TEST CASES

Module: RECOMMEND | Tutor Recommendations + Learning Path Recommendations | Maps to: F10, F11

TC ID	Test Category	Test Scenario	Precondition	Test Steps	Expected Result	Actual Result	Req. Mapped	Priority	Status	Defect ID
TC-REC-01	Functional	Tutor recommendations shown based on booking history	Student has ≥1 completed bookings in 'Math'	1. Log in as student 2. Go to 'Recommended for You' section	Tutors related to past booking subjects (e.g., Math) displayed as recommendations	Tutors related to past booking subjects displayed	F10	P1 – Critical	Pass	
TC-REC-02	Negative	Recommendations for new user with no history	Student registered but has no bookings	1. Log in as new student 2. Go to Recommendations section	Default popular tutors shown OR message: 'Book your first session to get personalized suggestions'	Default tutors shown	F10	P2 – High	Pass	
TC-REC-03	Integration	Recommendation updates after new booking	Student completes booking in a new subject	1. Student completes a 'Physics' booking (first) 2. Revisit Recommendations section	Physics tutors now appear in recommendations list	Physics tutors now appear in recommendations	F10	P1 – Critical	Pass	
TC-REC-04	Functional	Learning path recommendations returned for subject	Student searches or books 'Data Structures'	1. Student books/ views 'Data Structures' content 2. Go to Learning Path tab	Ordered learning path displayed: e.g., Arrays → Linked Lists → Trees → Graphs	Ordered learning path displayed	F11	P1 – Critical	Pass	
TC-REC-05	Negative	Recommendation API returns error — graceful fallback	Recommendation service is unavailable (simulated)	1. Disable recommendation module 2. Student navigates to Recommendations section	UI shows fallback message: 'Recommendations unavailable. Showing popular tutors.' No crash.	Fallback message shown; no crash	F10, F11	P1 – Critical	Pass	
TC-REC-06	Boundary	Student with exactly 1 booking receives recommendations	Student has exactly 1 completed booking	1. Ensure 1 completed booking exists 2. Go to Recommendations section	At least 1 relevant tutor recommendation shown	At least 1 relevant recommendation shown	F10	P2 – High	Pass	
TC-REC-07	Integration	Learning path aligns with available tutors in system	Learning path for 'Machine Learning' generated	1. View learning path for Machine Learning 2. Check that tutors exist for recommended subjects	Each step in learning path has at least one associated tutor available	Each step has at least one tutor available	F10, F11	P2 – High	Pass	
TC-REC-08	Negative	Unknown subject – no learning path exists	Student requests learning path for 'Astrology'	1. Manually navigate to / learning-path?subject=Astrology	Message: 'No learning path available for this subject'	Message: 'No learning path available'	F11	P2 – High	Pass	
TC-REC-09	Security	Unauthenticated access to recommendation API	User not logged in	1. Send GET /api/ recommendations without auth token	API returns 401 Unauthorized	API returns 401 Unauthorized	F10	P1 – Critical	Pass	
TC-REC-10	Role-Based	Tutor does not see student-facing recommendations	Tutor logged in	1. Log in as tutor 2. Check navigation/ dashboard for Recommendations tab	No recommendations section visible in tutor dashboard	No recommendations section visible	F10	P2 – High	Pass	

TC-REC-11	Functional	Recommendation respects tutor rating threshold	Student has booking history	1. Go to Recommendations section 2. Inspect all recommended tutors	All recommended tutors have rating ≥ configured minimum threshold	All recommended tutors meet minimum rating	F10	P2 – High	Pass	
Total Test Cases in this Sheet: 11										